Maxwell from the Unified 4D Toy Model: Localized 5D Gauge Dynamics, Brane Reduction, and KK Corrections

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Abstract

We derive brane-effective electromagnetism from a localized $4+1$ Maxwell sector embedded in the unified toy model. The gauge field is a bulk potential A_M with kinetic term $-Z(w)F_{MN}F^{MN}/(4\mu_0)$, where the transverse profile $Z(w)$ localizes gauge dynamics near the brane at $w=0$. Varying the action yields the localized Maxwell equations $\partial_M(ZF^{MN}) + \xi^{-1}\partial^N(\partial\cdot A) = \mu_0 J^N$, the Bianchi identities, and an exact divergence consistency identity linking the gauge condition to current conservation. For brane-localized conserved sources, axial gauge, and a brane-dominant zero mode, integrating over w reduces the theory to standard $3+1$ Maxwell with an effective coupling $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$, where $Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w)\,dw$. For a Gaussian profile $Z(w) = e^{-w^2/\lambda^2}$, $\mu_0^{\text{eff}} = \mu_0/(\lambda\sqrt{\pi})$ and the transverse Sturm–Liouville problem yields a discrete KK tower with masses $m_n^2 = 2n/\lambda^2$ and a brane parity selection rule. The resulting static brane potential is Coulomb plus Yukawa-suppressed corrections with a fixed coefficient pattern; the leading deviation scales as $\frac{1}{2}e^{-2r/\lambda}$. For time-dependent sources, the brane-to-brane response depends only on the Lorentz scalar k^2 (with the retarded prescription), ensuring brane Lorentz covariance; massive modes add causal tail terms inside the light cone. Finally, a minimally coupled brane matter model yields a conserved Noether current and an off-shell identity that supplies the Maxwell source consistency condition. In the updated charge ontology, a fixed defect branch carries $q_\star = \eta_Q e_\star$ with $\eta_Q = \pm 1$, while canonical normalization of the reduced zero mode gives $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$; circulation is not used to define electric charge, and the zero-mode Maxwell limit suppresses the mixed $(A_w, J^w, F_{\mu w})$ sector only as a controlled far-field approximation. All principal identities and reductions are backed by a compact Wolfram Language referee suite.

1 Introduction

In the unified toy model developed across the preceding papers, the vacuum is treated as a dynamical medium with a brane/bulk decomposition, and matter-like excitations are associated with localized defect structures. Paper VII introduced a unified $4+1$ ("5D") formulation that cleanly separates *exact* bulk field equations from *controlled* brane-effective reductions. The present paper isolates and completes the electromagnetic sector of that framework: we show that standard $3+1$ Maxwell theory on the brane arises as a controlled long-distance/low-frequency limit of a localized $4+1$ Maxwell action, and we quantify the leading, falsifiable deviations when transverse excitations are relevant.

Earlier in the series, electromagnetism was developed through an "EM-from-flow" dictionary: with appropriate variable identifications, the homogeneous Maxwell equations can be realized as kinematic identities and the Lorentz-force structure can be motivated from a gauged phase-gradient (Madelung) description of matter. While that viewpoint is useful for building intuition and for organizing brane-observed fields, it leaves open a question that referees naturally insist upon: *where do the inhomogeneous Maxwell equations and the correct propagator structure come* *from, without postulating them?* The purpose of this paper is to supply that missing dynamical step within the unified $4+1$ model.

In the charge ontology adopted after Paper VII, this paper no longer uses circulation, throat radius, or breathing variables to define electric charge. Instead, the electric branch sign is a fixed topological puncture orientation $\eta_Q = \pm 1$, the microscopic defect-branch coupling is $q_\star \equiv \eta_Q e_\star$ with $e_\star > 0$, and the observable brane strength is controlled by the localization thickness through $Z_{\text{int}} = \int Z(w)\,\mathrm{d}w$ (equivalently by canonical zero-mode normalization, $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$). The familiar zero-mode Maxwell reduction is therefore a controlled far-field brane law: it suppresses the mixed core channels $(A_w, J^w, F_{\mu w})$ without erasing them from the microscopic ontology.

The key ingredient is a genuine $4+1$ gauge potential A_M with a localized Maxwell kinetic term,

$$-\frac{Z(w)}{4\mu_0} F_{MN} F^{MN},$$

where the nonnegative profile $Z(w)$ confines the gauge dynamics near the brane at $w = 0$. For concreteness (and analytic control) we adopt a Gaussian profile $Z(w) = \exp(-w^2/\lambda^2)$, but the logical structure applies to any profile with finite integral $Z_{\text{int}} = \int Z(w)\, dw$. We show:

- **Bulk dynamics (exact):** Varying the localized action yields the localized $4+1$ Maxwell equations $\partial_M(ZF^{MN}) + (1/\xi)\partial^N(\partial\cdot A) = \mu_0 J^N$, together with exact Bianchi identities and a divergence consistency identity linking the gauge condition to current conservation.

- **Brane Maxwell limit (controlled):** For brane-localized sources, axial gauge, and a brane-dominant zero mode, integrating over the transverse direction produces standard $3+1$ Maxwell theory with an effective coupling

$$\mu_0^{\text{eff}} = \frac{\mu_0}{Z_{\text{int}}} \quad \Rightarrow \quad \mu_0^{\text{eff}} = \frac{\mu_0}{\lambda\sqrt{\pi}} \ \text{ for Gaussian } Z.$$

  The same integrated equation makes explicit the correction terms that appear when w-dependence (KK excitations) is not negligible.

- **KK corrections (derived, quantitative):** For Gaussian localization, the transverse Sturm–Liouville problem can be solved in closed form. The resulting discrete KK tower yields Yukawa-suppressed corrections to Coulomb's law with a fixed coefficient pattern; the leading correction scales as $\frac{1}{2}e^{-2r/\lambda}$ relative to the Coulomb term.

- **Moving sources (covariant, causal):** The effective brane response is a function of the brane Lorentz scalar k^2 (with the standard retarded prescription), implying brane Lorentz covariance for moving sources. Mode-by-mode retarded Green functions make the causal structure explicit: massive modes add tail-type response inside the light cone, while remaining fully causal.

- **Source consistency (derived, not assumed):** A minimally coupled brane matter model has a $U(1)$ phase symmetry whose Noether current provides a conserved source. This directly supplies the conservation condition required by the Maxwell divergence identity, closing the logical loop between gauge symmetry, matter dynamics, and Maxwell consistency.

Exact vs controlled vs regime statements. To keep the logical status of each step unambiguous, we adopt the following taxonomy throughout: (i) *exact* statements are identities or Euler–Lagrange equations obtained directly from the localized action (e.g. Bianchi identities, the divergence identity); (ii) *controlled* statements require explicit reduction assumptions (e.g. axial gauge plus a brane-dominant zero mode); and (iii) *regime* statements interpret the controlled limit in terms of observable scales (e.g. $r \gg \lambda$, $\omega \ll 2/\lambda$) and summarize the size of corrections.

Reproducibility and referee suite. All core equalities and identities in this paper are accompanied by a surgical Wolfram Language verification suite that checks (a) the Euler–Lagrange equations component-by-component, (b) the Bianchi identities, (c) the divergence consistency identity, (d) the controlled brane reduction and the effective coupling μ_0^{eff} together with the equivalent canonical relation $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$, (e) the Gaussian KK spectrum and brane couplings, (f) the k^2-only dependence of the effective brane propagator for a fixed defect branch, and (g) the matter Noether current and off-shell identity with $J^\mu_\psi = q_\star j^\mu$. A concise summary of these scripts and what each establishes is provided in section A.

Roadmap. Section 2 fixes conventions and defines the localized action. Section 3 derives the bulk Euler–Lagrange equations and the consistency identities. Section 4 records the gauge-variation structure and brane Lorentz covariance. Section 5 presents the controlled brane reduction and derives μ_0^{eff}. Section 6 solves the Gaussian KK problem and derives the Coulomb plus Yukawa corrections. Section 7 extends the discussion to moving sources and retarded structure. Section 8 derives matter current conservation and links it to Maxwell consistency. Finally, Section 9 summarizes the validity regime, leading deviations, and falsifiable tests.

2 Conventions and localized 5D Maxwell action

This paper isolates the gauge-field sector of the unified model in a standard 4+1 ("5D") spacetime notation. Concretely, we work with one time coordinate, three brane spatial coordinates, and one transverse bulk coordinate w. This is the same localization sector used inside the full unified paper, but here we treat it as a self-contained dynamical system whose controlled brane limit reproduces ordinary $3+1$ Maxwell theory (with computable corrections).

2.1 Coordinates, indices, and metric

We use bulk coordinates

$$x^M = (t, x, y, z, w), \qquad M, N, \ldots \in \{0, 1, 2, 3, 4\}, \tag{1}$$

with the brane identified as the hypersurface $w = 0$. Greek indices $\mu, \nu, \ldots \in \{0, 1, 2, 3\}$ refer to brane spacetime coordinates $x^\mu = (t, x, y, z)$, and Latin indices $a, b, \ldots \in \{1, 2, 3\}$ refer to the brane spatial coordinates (x, y, z).

We take a flat bulk metric with "mostly plus" signature,

$$\eta_{MN} = \text{diag}(-1, 1, 1, 1, 1), \qquad \eta^{MN} = (\eta_{MN})^{-1}. \tag{2}$$

Indices are raised/lowered with η_{MN} and η^{MN}. Derivatives are $\partial_M \equiv \partial/\partial x^M$, and we define the $4+1$ d'Alembertian

$$\Box \equiv \eta^{MN}\partial_M\partial_N = -\partial_t^2 + \nabla^2 + \partial_w^2, \tag{3}$$

where ∇ is the ordinary gradient in the brane spatial directions (x, y, z). Throughout, we work in units where the brane light-cone speed is unity; factors of c can be restored by $t \mapsto ct$ in the metric and $\partial_t \mapsto c^{-1}\partial_t$.

2.2 Profile and normalization

The hallmark of the localization mechanism is a nonnegative profile $Z(w)$ multiplying the Maxwell kinetic term. In this paper we adopt a Gaussian profile,

$$Z(w) = \exp\left(-\frac{w^2}{\lambda^2}\right), \tag{4}$$

with confinement width $\lambda > 0$. Its integral,

$$Z_{\text{int}} \equiv \int_{-\infty}^{\infty} Z(w)\,\mathrm{d}w = \lambda\sqrt{\pi}, \tag{5}$$

is finite, which is the key condition ensuring a normalizable zero mode and an effective $3+1$ gauge coupling on the brane.

All integrations by parts in w are taken with the standard assumption that the relevant boundary terms vanish as $|w| \to \infty$. For Gaussian $Z(w)$, it is sufficient that $Z(w)F^{wN}$ and $Z(w)A_M\partial_w(\cdots)$ decay rapidly enough; this holds for the localized configurations considered below and is explicitly verified in the symbolic checks by working with smooth test fields and then taking controlled limits.

2.3 Fields, field strength, and gauge fixing

The bulk gauge potential is a one-form (covector field) $A_M(x)$. Its field strength is

$$F_{MN} \equiv \partial_M A_N - \partial_N A_M, \qquad F^{MN} \equiv \eta^{MP}\eta^{NQ}F_{PQ}. \tag{6}$$

We also use the covariant divergence

$$\partial\cdot A \equiv \partial_M A^M, \qquad A^M \equiv \eta^{MN}A_N. \tag{7}$$

To make the kinetic operator invertible and to streamline the consistency checks, we include a covariant gauge-fixing term with parameter ξ. This is a standard device: ξ labels a family of equivalent descriptions, and all gauge-invariant observables are ξ-independent provided the source current is conserved. We keep ξ arbitrary until we specialize to Lorenz gauge ($\partial\cdot A = 0$) when convenient.

2.4 Action and source bookkeeping

The localized Maxwell sector is defined by the action

$$S_{\text{EM}}[A;Z] = \int \mathrm{d}t\,\mathrm{d}^3x\,\mathrm{d}w\ \left[-\frac{Z(w)}{4\mu_0}\,F_{MN}F^{MN} - \frac{1}{2\xi\mu_0}\,(\partial\cdot A)^2 - A_M J^M\right]. \tag{8}$$

The source $J^M(x)$ is taken to be a contravariant current (a vector field), and the interaction is fixed to the sign convention $\mathcal{L}_{\text{int}} = -A_M J^M$. This particular bookkeeping choice is important: it ensures that under a gauge variation $\delta A_M = \partial_M\chi$, the interaction term varies as $\delta\mathcal{L}_{\text{int}} = -\partial_M(\chi J^M) + \chi\,\partial_M J^M$, making explicit the equivalence between gauge invariance of the source coupling and current conservation (up to boundary terms). We emphasize this convention here because it was historically the dominant source of component/sign mismatches in symbolic implementations.

When embedded into the unified model, J^M should be read as a shorthand total source,

$$J^M_{\text{tot}} = J^M_\psi + J^M_{\text{ext}}. \tag{9}$$

In the fully coupled action, the explicit Maxwell source term is therefore $-A_M J^M_{\text{ext}}$, while the matter current J^M_ψ is generated separately by minimal coupling and must not be double-counted. In the present Maxwell-only derivation we keep the compact placeholder J^M, but in the coupled reading it is always understood as J^M_{tot}. Neutralizing backgrounds or jellium subtractions belong in J^0_{ext}, not in a fluctuating electric-charge dictionary.

Finally, we note a transformation-type convention that will matter later: A_μ transforms as a covector under brane Lorentz transformations, while J^μ transforms as a vector. With these choices, the scalar contraction $A_\mu J^\mu$ and the invariant $F_{\mu\nu}F^{\mu\nu}$ are manifestly brane-Lorentz invariant in the localized limit, as verified in the accompanying symbolic suite.

3 Euler–Lagrange equations and consistency identities

This section records the exact field equations and identities implied by the localized Maxwell action equation (8). The accompanying Wolfram Language referee suite verifies these results both in compact index form and component-by-component for $N \in \{t, x, y, z, w\}$, including the sign conventions and transformation types stated in section 2.

3.1 Field equations from variation

Varying A_N in equation (8) gives the Euler–Lagrange equations. Using

$$\delta F_{MN} = \partial_M(\delta A_N) - \partial_N(\delta A_M), \tag{10}$$

the variation of the kinetic term can be written as

$$\delta\Big(-\frac{Z}{4\mu_0}F_{MN}F^{MN}\Big) = -\frac{Z}{2\mu_0}F^{MN}\delta F_{MN} = -\frac{Z}{\mu_0}F^{MN}\partial_M(\delta A_N), \tag{11}$$

where antisymmetry of F^{MN} was used in the last step. Integrating by parts (and dropping boundary terms, justified in section 2.2) gives

$$\delta S_{\text{kin}} = \int \mathrm{d}^5x\, \frac{1}{\mu_0}\,\partial_M(Z\,F^{MN})\,\delta A_N. \tag{12}$$

Similarly, the gauge-fixing variation is

$$\delta\Big(-\frac{1}{2\xi\mu_0}(\partial\cdot A)^2\Big) = -\frac{1}{\xi\mu_0}(\partial\cdot A)\,\partial_M(\delta A^M) \;\;\Rightarrow\;\; \delta S_{\text{gf}} = \int \mathrm{d}^5x\,\frac{1}{\xi\mu_0}\,\partial^N(\partial\cdot A)\,\delta A_N, \tag{13}$$

where we used $\delta A^M = \eta^{MN}\delta A_N$ and integrated by parts. Finally, the source term varies as

$$\delta(-A_M J^M) = -J^N\,\delta A_N. \tag{14}$$

Requiring $\delta S_{\text{EM}} = 0$ for arbitrary δA_N yields the localized $4+1$ Maxwell equations,

$$\partial_M\Big(Z(w)F^{MN}\Big) + \frac{1}{\xi}\,\partial^N(\partial\cdot A) = \mu_0\,J^N. \tag{15}$$

We will refer to the left-hand side as the Maxwell operator,

$$\mathcal{M}^N[A] \equiv \partial_M\Big(ZF^{MN}\Big) + \frac{1}{\xi}\,\partial^N(\partial\cdot A), \qquad \mathcal{M}^N[A] = \mu_0\,J^N. \tag{16}$$

3.2 Bianchi identities (homogeneous Maxwell)

The definition equation (6) implies the Bianchi identities,

$$\partial_{[L}F_{MN]} = 0, \tag{17}$$

or explicitly,

$$\partial_L F_{MN} + \partial_M F_{NL} + \partial_N F_{LM} = 0. \tag{18}$$

These are exact, off-shell identities (they hold for any smooth A_M), and constitute the homogeneous Maxwell equations upon projection to brane observables in the controlled reduction discussed later.

3.3 Divergence identity and current conservation

Taking ∂_N of equation (15) yields a consistency identity that ties the gauge condition to current conservation. The key point is that the double divergence of the antisymmetric tensor ZF^{MN} vanishes:

$$\partial_N\partial_M(ZF^{MN}) = -\partial_M\partial_N(ZF^{MN}) \quad\Rightarrow\quad \partial_N\partial_M(ZF^{MN}) = 0, \tag{19}$$

where we relabeled dummy indices $M \leftrightarrow N$ and used commutativity of partial derivatives in flat space. Therefore,

$$\partial_N\left(\mathcal{M}^N[A] - \mu_0\, J^N\right) = \frac{1}{\xi}\,\partial_N\partial^N(\partial\cdot A) - \mu_0\,\partial_N J^N = \frac{1}{\xi}\,\Box(\partial\cdot A) - \mu_0\,\partial_N J^N. \tag{20}$$

In particular, in Lorenz gauge $\partial\cdot A = 0$ the Maxwell equations enforce $4+1$ current conservation,

$$\partial_N J^N = 0. \tag{21}$$

Conversely, for a conserved source $\partial_N J^N = 0$, the gauge-fixing term $\propto (\partial\cdot A)^2$ can be viewed as selecting a representative within the gauge orbit without affecting gauge-invariant observables. This identity is the precise statement of "Maxwell consistency" in the localized $4+1$ setting and will later be matched to the independently derived matter Noether conservation law in section 8.

4 Gauge structure and brane Lorentz covariance

The localized Maxwell sector has the usual gauge structure in the bulk, modified only by the presence of a gauge-fixing term (included here for operator invertibility and for clean symbolic verification). In addition, because the localization profile $Z(w)$ depends only on the transverse coordinate w, the theory preserves exact $3+1$ Lorentz symmetry along the brane directions. This section records the key structural facts that will be used later in the brane reduction and propagator discussion.

4.1 Gauge variation and boundary terms

Consider an infinitesimal gauge transformation generated by a scalar function $\chi(x)$,

$$\delta_\chi A_M = \partial_M\chi. \tag{22}$$

The field strength is gauge invariant,

$$\delta_\chi F_{MN} = \partial_M\partial_N\chi - \partial_N\partial_M\chi = 0, \tag{23}$$

so the localized kinetic term $-Z\,F^2/(4\mu_0)$ is unchanged. The source interaction varies as

$$\delta_\chi\mathcal{L}_{\text{int}} = -(\delta_\chi A_M)J^M = -(\partial_M\chi)J^M = -\partial_M(\chi J^M) + \chi\,\partial_M J^M. \tag{24}$$

Thus, up to a total derivative, gauge invariance of the source coupling is equivalent to current conservation. In particular, if $\partial_M J^M = 0$ and boundary terms can be dropped, then $\delta_\chi S_{\text{int}} = 0$.

The gauge-fixing term explicitly breaks gauge invariance in the standard way. Using $\partial\cdot A = \partial_M A^M$, we have

$$\delta_\chi(\partial\cdot A) = \partial_M\delta_\chi A^M = \partial_M\partial^M\chi = \Box\chi, \tag{25}$$

so $(\partial\cdot A)^2$ is not invariant. This is expected: the gauge-fixing term is introduced only to render the kinetic operator invertible, and physical (gauge-invariant) observables are independent of the choice of ξ when the source is conserved. In our referee suite we keep the gauge-fixing term because it makes the operator comparison and divergence identity checks unambiguous.

4.2 Transformation types: covector vs vector

A recurring source of confusion in component-based checks is the difference between how the gauge potential A_μ and the current J^μ transform under Lorentz transformations. In the present conventions:

- The potential A_μ is a *covector* (one-form) on the brane.
- The current J^μ is a *vector* on the brane.

Let $\Lambda^\mu{}_\nu$ be a $3+1$ Lorentz transformation acting on brane coordinates $x'^\mu = \Lambda^\mu{}_\nu x^\nu$. Then the fields transform as

$$A'_\mu(x') = (\Lambda^{-1})^\nu{}_\mu \, A_\nu(x), \qquad J'^\mu(x') = \Lambda^\mu{}_\nu \, J^\nu(x). \tag{26}$$

With these assignments, the contraction $A_\mu J^\mu$ is a scalar:

$$A'_\mu(x')J'^\mu(x') = (\Lambda^{-1})^\nu{}_\mu A_\nu(x)\, \Lambda^\mu{}_\rho J^\rho(x) = \delta^\nu{}_\rho \, A_\nu(x) J^\rho(x) = A_\mu(x) J^\mu(x). \tag{27}$$

Likewise, the brane components $F_{\mu\nu}$ transform as a rank-2 tensor and the invariant $F_{\mu\nu}F^{\mu\nu}$ is unchanged. These transformation-type assignments are implemented explicitly in the symbolic referee suite and are the reason invariance checks succeed without introducing spurious sign or index mismatches.

4.3 Effective brane propagator depends only on k^2

Because $Z(w)$ depends only on the transverse coordinate w, the localized theory is exactly invariant under the $3+1$ Poincaré group acting along the brane. In particular, after Fourier transforming in the brane coordinates, $\partial_\mu \to \mathrm{i}k_\mu$, the brane momentum enters only through the scalar

$$k^2 \equiv \eta^{\mu\nu} k_\mu k_\nu = -\omega^2 + \mathbf{k}^2, \tag{28}$$

which is invariant under brane Lorentz transformations. The KK decomposition associated with the Sturm–Liouville problem in w (worked out explicitly for Gaussian Z in section 6) yields an effective brane-to-brane propagator of the schematic form

$$D_{\text{eff}}(k^2) = \sum_{n\geq 0} \frac{c_n}{k^2 + m_n^2 + \mathrm{i}\epsilon}, \tag{29}$$

where m_n are the KK masses and c_n are the brane couplings determined by the mode profiles at $w=0$. Since the right-hand side depends on the brane four-momentum only through k^2, it is manifestly brane-Lorentz covariant. This has two important consequences:

1. *Static limit:* D_{eff} reproduces a Coulomb potential from the massless mode plus Yukawa-suppressed corrections from massive KK modes (quantified in section 6.3).
2. *Moving sources:* each KK mode propagates as a covariant massless or massive wave equation in $3+1$, so retarded solutions for moving defect branches retain the correct Lorentz structure mode-by-mode; summing the tower preserves this covariance.

In later sections we will use equation (29) both as a compact organizing principle and as a bridge between the exact 5D action and the controlled 3+1 effective description.

5 Controlled brane reduction and $\mu_{0,\text{eff}}$

The $4+1$ equations equation (15) describe a gauge field living in the bulk with a kinetic term localized near the brane by the profile $Z(w)$. In this section we explain (i) how axial gauge cleanly removes A_w while preserving an ordinary $3+1$ residual gauge freedom, (ii) how brane-localized sources are implemented, and (iii) how the integrated equations reduce to standard $3+1$ Maxwell with an effective coupling $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$. The accompanying referee suite verifies each step symbolically, including the consistency of the $N = w$ component under the reduction assumptions.

5.1 Axial gauge reachability and residual $3+1$ freedom

Start from a general configuration $A_M(x^\mu, w)$. Under a gauge transformation $\delta A_M = \partial_M \chi$, the transverse component transforms as $A_w \mapsto A_w + \partial_w \chi$. Provided A_w is sufficiently regular in w, we may reach axial gauge $A_w = 0$ by choosing

$$\chi_{\text{Ax}}(x^\mu, w) \equiv -\int_0^w A_w(x^\mu, w')\, \mathrm{d}w', \tag{30}$$

so that $\partial_w \chi_{\text{Ax}} = -A_w$. In axial gauge the residual gauge freedom is characterized by gauge parameters satisfying $\partial_w \chi = 0$, i.e.

$$\chi(x^\mu, w) = \chi_b(x^\mu), \tag{31}$$

which is precisely the usual $3+1$ gauge freedom on the brane.

With $A_w = 0$, the full divergence simplifies to

$$\partial\cdot A = \partial_M A^M = \partial_\mu A^\mu + \partial_w A^w = \partial_\mu A^\mu, \tag{32}$$

so in axial gauge one has $\partial_M A^M = \partial_\mu A^\mu$. In the controlled zero-mode reduction discussed below, the clean brane Maxwell law is obtained by imposing Lorenz gauge before integrating over w, or equivalently by fixing gauge after passing to the reduced $3+1$ theory.

5.2 Brane-localized sources

To model a conserved source confined to the brane in the controlled far-field Maxwell reduction, we take

$$J^\mu(x^\nu, w) = J_b^\mu(x^\nu)\, \delta(w), \qquad J^w(x^\nu, w) = 0, \tag{33}$$

where J_b^μ is a $3+1$ current on the brane. The condition $J^w = 0$ is therefore part of the reduction ansatz used to isolate the brane Maxwell sector: it expresses the absence of net transverse current in that limit and is the condition required by consistency of the $N = w$ Maxwell equation under the axial/zero-mode reduction (see below). It should not be read as a microscopic claim that the charged core lacks mixed-sector structure. Current conservation $\partial_M J^M = 0$ then reduces to the usual brane condition $\partial_\mu J_b^\mu = 0$.

In the coupled defect reading, J_b^μ may represent the total brane source ${J_{\text{tot}}}_b^\mu = {J_\psi}_b^\mu + {J_{\text{ext}}}_b^\mu$. The sign of the dynamical electric branch is then carried separately by η_Q through ${J_\psi}_b^\mu = q_\star j_b^\mu$, while any neutralizing background resides in ${J_{\text{ext}}}_b^0$.

For later convenience we also define the w-integrated current

$$J_{\text{eff}}^\mu(x) \equiv \int_{-\infty}^{\infty} J^\mu(x, w)\, \mathrm{d}w, \tag{34}$$

so that equation (33) implies $J_{\text{eff}}^\mu = J_b^\mu$.

5.3 Integrated EOM and effective coupling

Write the ν-components of the Maxwell equation equation (15) as

$$\partial_M\Big(ZF^{M\nu}\Big)+\frac{1}{\xi}\,\partial^\nu(\partial\cdot A)=\mu_0\,J^\nu,\qquad \nu\in\{0,1,2,3\}. \tag{35}$$

Integrating over w gives the exact identity

$$\partial_\mu\Big(\int Z\,F^{\mu\nu}\,\mathrm{d}w\Big)+\underbrace{\int\partial_w(Z\,F^{w\nu})\,\mathrm{d}w}_{[\,Z\,F^{w\nu}\,]_{-\infty}^{+\infty}}+\frac{1}{\xi}\,\partial^\nu\Big(\int(\partial\cdot A)\,\mathrm{d}w\Big)=\mu_0\,J^\nu_{\text{eff}}(x), \tag{36}$$

where we used $Z=Z(w)$ and commuted ∂_μ past the w-integral. For localized field configurations the boundary term $[\,Z\,F^{w\nu}\,]_{-\infty}^{+\infty}$ vanishes. It is then natural to define a weighted brane field strength by

$$\bar F^{\mu\nu}(x)\equiv\frac{1}{Z_{\text{int}}}\int_{-\infty}^{\infty}Z(w)\,F^{\mu\nu}(x,w)\,\mathrm{d}w, \tag{37}$$

so that equation (36) becomes

$$\partial_\mu\bar F^{\mu\nu}+\frac{1}{\xi\,Z_{\text{int}}}\,\partial^\nu\Big(\int(\partial\cdot A)\,\mathrm{d}w\Big)=\frac{\mu_0}{Z_{\text{int}}}\,J^\nu_{\text{eff}}. \tag{38}$$

Equation equation (38) is an *exact* integrated brane equation, but it is not yet standard Maxwell because $\bar F^{\mu\nu}$ is a weighted bulk average and the gauge-fixing term retains an explicit w-integral.

To obtain the familiar $3+1$ Maxwell form we now take a controlled zero-mode limit: in axial gauge $A_w=0$, assume the brane components are w-independent over the localization region,

$$A_\mu(x,w)\simeq a_\mu(x),\qquad \partial_wA_\mu\simeq 0, \tag{39}$$

so that $F^{w\nu}=-\partial^wA^\nu\simeq 0$ and $F^{\mu\nu}(x,w)\simeq f^{\mu\nu}(x)$, where

$$f_{\mu\nu}(x)\equiv\partial_\mu a_\nu-\partial_\nu a_\mu. \tag{40}$$

Then equation (37) gives $\bar F^{\mu\nu}\simeq f^{\mu\nu}$. However, because the five-dimensional gauge-fixing term is not weighted by $Z(w)$, the integrated gauge-fixing contribution in equation (38) is not naturally proportional to Z_{int}. The clean zero-mode reduction is therefore obtained by imposing Lorenz gauge on the five-dimensional representative before reduction, so that $\partial\cdot A=0$ and the explicit gauge-fixing term in equation (38) vanishes identically. Under these assumptions, equation (38) reduces to

$$\partial_\mu f^{\mu\nu}=\mu_0^{\text{eff}}\,J^\nu_{\text{eff}},\qquad \mu_0^{\text{eff}}\equiv\frac{\mu_0}{Z_{\text{int}}}. \tag{41}$$

For a brane-localized current equation (33), $J^\nu_{\text{eff}}=J^\nu_b$, and using equation (5) we obtain the effective coupling

$$\mu_0^{\text{eff}}=\frac{\mu_0}{\lambda\sqrt{\pi}}. \tag{42}$$

A $3+1$ gauge-fixed representative may then be chosen after reduction, for example $\partial_\mu a^\mu=0$; if one wants an explicit gauge-fixing term in the reduced theory, it should be introduced at the $3+1$ level rather than inferred directly from the unweighted five-dimensional term.

Consistency of the w-component. The transverse component of equation (15) reads

$$\partial_M\Big(ZF^{Mw}\Big)+\frac{1}{\xi}\,\partial^w(\partial\cdot A)=\mu_0\,J^w. \tag{43}$$

In axial gauge $A_w=0$ with the zero-mode ansatz equation (39), we have $F^{\mu w}=-\partial^wA^\mu\simeq 0$ and $\partial^w(\partial\cdot A)\simeq 0$, so equation (43) reduces to $J^w\simeq 0$. Thus the controlled brane limit is internally consistent precisely when there is no net transverse current, matching the source model equation (33).

Correction terms and regime of validity. Equation equation (38) makes clear what is being assumed in the reduction. Deviations from standard $3+1$ Maxwell arise when: (i) A_μ has significant w-dependence (excited KK modes), so that $\bar{F}^{\mu\nu} \neq f^{\mu\nu}$; (ii) $F^{w\nu}$ does not decay rapidly enough to drop the boundary term in equation (36); or (iii) $J^w \neq 0$. For Gaussian localization, the KK tower and its brane couplings can be computed explicitly, yielding Yukawa-suppressed corrections to Coulomb and to radiation propagation; we quantify these effects in section 6 and discuss their interpretation for moving sources in section 7.

Far-field scope of the zero-mode limit. The assumptions $A_w = 0$, $\partial_w A_\mu \simeq 0$, and $J^w \simeq 0$ define a controlled *far-field brane Maxwell limit.* Near a topologically charged defect core one generically expects nontrivial mixed-sector structure in $(A_w, J^w, F_{\mu w})$. Those channels are precisely where the puncture orientation η_Q and any future mixed-twist/spin microphysics would live. Accordingly, the reduced brane Maxwell theory derived here is the effective far-field law seen on the brane, not a statement that the microscopic charged defect lacks odd-w or mixed core structure.

5.4 Canonical normalization and thickness-controlled brane charge

The reduction result $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$ is the most compact way to state the brane Maxwell limit, but the same physics can be expressed as a thickness-controlled charge normalization. Under the zero-mode ansatz, the effective brane action takes the schematic form

$$S_{\text{EM}}^{\text{eff}} \simeq \int \mathrm{d}^4x \left[-\frac{Z_{\text{int}}}{4\mu_0} f_{\mu\nu} f^{\mu\nu} - a_\mu J_b^\mu \right]. \tag{44}$$

For a fixed defect branch with localized matter current $J_{\psi}{}_b^\mu = q_\star j_b^\mu$, canonically normalizing the brane zero mode by

$$a_\mu = \frac{a_\mu^{\text{can}}}{\sqrt{Z_{\text{int}}}} \tag{45}$$

gives

$$S_{\text{EM}}^{\text{eff}} \simeq \int \mathrm{d}^4x \left[-\frac{1}{4\mu_0} f_{\mu\nu}^{\text{can}} f_{\text{can}}^{\mu\nu} - a_\mu^{\text{can}} \frac{J_b^\mu}{\sqrt{Z_{\text{int}}}} \right]. \tag{46}$$

Thus the canonically normalized brane coupling of a fixed defect branch is

$$q_{\text{eff}} = \frac{q_\star}{\sqrt{Z_{\text{int}}}} = \eta_Q\, e_{\text{eff}}, \qquad e_{\text{eff}} \equiv \frac{e_\star}{\sqrt{Z_{\text{int}}}}. \tag{47}$$

The microscopic branch label $q_\star$ itself remains fixed; only the reduced brane coupling strength scales with the localization thickness. For the Gaussian profile $Z(w) = \exp(-w^2/\lambda^2)$, one has

$$Z_{\text{int}} = \sqrt{\pi}\,\lambda, \qquad e_{\text{eff}} \propto \lambda^{-1/2}, \tag{48}$$

so thicker localization weakens the observable brane electric coupling.

6 Gaussian KK tower and corrections to Coulomb

For a finite localization integral Z_{int}, the localized Maxwell sector admits a normalizable zero mode that reproduces ordinary $3+1$ Maxwell on the brane, plus a discrete tower of massive excitations (KK modes) whose exchange produces Yukawa-suppressed corrections. For the Gaussian profile $Z(w) = e^{-w^2/\lambda^2}$ these structures can be worked out in closed form. In this section we summarize: (i) the Sturm–Liouville eigenproblem in the transverse coordinate w, (ii) the resulting spectrum and mode normalization, (iii) the brane couplings and parity selection rule, and (iv) the resulting static potential (Coulomb plus Yukawa corrections). These results are verified in the accompanying Wolfram Language suite.

6.1 Sturm–Liouville problem and spectrum

Work in axial gauge $A_w = 0$ and, for definiteness, in Lorenz gauge for the brane components. In the source-free region, the brane components of equation (15) separate into a brane wave operator and a transverse operator acting on the w-profile. This motivates a mode expansion

$$A_\mu(x, w) = \sum_{n\geq 0} a_\mu^{(n)}(x)\, f_n(w), \tag{49}$$

where the $f_n(w)$ are eigenfunctions of the transverse Sturm–Liouville operator associated with the weighted Maxwell kinetic term:

$$-\frac{\mathrm{d}}{\mathrm{d}w}\left(Z(w)\,\frac{\mathrm{d}f_n}{\mathrm{d}w}\right) = m_n^2\, Z(w)\, f_n(w). \tag{50}$$

The weight is $Z(w)$, and normalizability is with respect to the inner product $\langle f, g\rangle \equiv \int_{-\infty}^{\infty} Z(w) f(w) g(w)\,\mathrm{d}w$.

For the Gaussian $Z(w) = \exp(-w^2/\lambda^2)$, writing $y \equiv w/\lambda$ turns equation (50) into

$$\frac{\mathrm{d}^2 f_n}{\mathrm{d}y^2} - 2y\,\frac{\mathrm{d}f_n}{\mathrm{d}y} + (m_n^2\lambda^2)\, f_n = 0, \tag{51}$$

which is the Hermite differential equation. A complete set of polynomial solutions is given by the physicists' Hermite polynomials,

$$f_n(w) = H_n\!\left(\frac{w}{\lambda}\right), \qquad m_n^2 = \frac{2n}{\lambda^2}, \qquad n = 0, 1, 2, \ldots \tag{52}$$

with $n = 0$ the massless mode. The massive modes correspond to $3+1$ Proca-type propagating degrees of freedom on the brane, with masses m_n.

6.2 Mode normalization and brane couplings

The natural norm induced by the action is

$$\|f_n\|^2 \equiv \int_{-\infty}^{\infty} Z(w)\, f_n(w)^2\,\mathrm{d}w. \tag{53}$$

For Gaussian Z and Hermite f_n, the norm is the standard Hermite orthogonality integral,

$$\|f_n\|^2 = \int_{-\infty}^{\infty} \exp\!\left(-\frac{w^2}{\lambda^2}\right) H_n\!\left(\frac{w}{\lambda}\right)^2 \mathrm{d}w = \lambda\sqrt{\pi}\, 2^n\, n!. \tag{54}$$

A brane-localized current $J^\mu(x, w) = J_b^\mu(x)\delta(w)$ excites each mode in proportion to its value at the brane. A convenient "brane coupling weight" is

$$c_n \equiv \frac{f_n(0)^2}{\|f_n\|^2}, \tag{55}$$

which will appear directly in the effective brane-to-brane propagator and in the static potential.

For Hermite polynomials, $H_n(0) = 0$ for odd n. Thus,

$$c_{2m+1} = 0, \qquad m = 0, 1, 2, \ldots \tag{56}$$

which is a simple parity selection rule: the profile $Z(w)$ is even in w, the brane sits at the symmetry point $w = 0$, and odd transverse modes do not couple to brane sources. This centered-source decoupling does not mean that odd-w core structure is absent microscopically; it means only that such structure can remain hidden from leading brane sources in the centered Maxwell reduction.

For even $n = 2m$, the closed-form value $H_{2m}(0) = (-1)^m (2m)!/m!$ gives

$$c_{2m} = \frac{H_{2m}(0)^2}{\lambda\sqrt{\pi}\, 2^{2m}(2m)!} = \frac{1}{\lambda\sqrt{\pi}}\frac{1}{4^m}\binom{2m}{m}. \tag{57}$$

In particular, the zero-mode coupling is

$$c_0 = \frac{1}{\lambda\sqrt{\pi}} = \frac{1}{Z_{\rm int}}, \tag{58}$$

which reproduces $\mu_0^{\rm eff} = \mu_0/Z_{\rm int}$ from equation (42).

6.3 Coulomb law and Yukawa corrections

The KK expansion equation (49) implies that brane momenta enter through the Lorentz scalar k^2, and the brane-to-brane propagator takes the form

$$D_{\rm eff}(k^2) = \sum_{n\geq 0} \frac{c_n}{k^2 + m_n^2 + i\epsilon}, \qquad m_n^2 = \frac{2n}{\lambda^2}. \tag{59}$$

With $\Box_4 = \partial_t^2 - \nabla^2$ and $k^2 = -\omega^2 + \mathbf{k}^2$, this denominator is the momentum-space partner of the mode operator $\Box_4 + m_n^2$ used below. In the static limit ($\omega = 0$), each contributing mode produces the standard three-dimensional Green's function for $\nabla^2 - m_n^2$, yielding a Yukawa potential for $m_n > 0$. For a fixed defect branch on the brane, modeled by $J_b^0 = q_\star\, \delta^{(3)}(\mathbf{x})$, the scalar potential on the brane is

$$A_0(r) = \frac{\mu_0 q_\star}{4\pi r} \sum_{n\geq 0} c_n\, e^{-m_n r}, \qquad r \equiv |\mathbf{x}|. \tag{60}$$

Separating the zero mode and using $\mu_0^{\rm eff} = \mu_0 c_0$, we can write

$$A_0(r) = \frac{\mu_0^{\rm eff} q_\star}{4\pi r}\left[1 + \sum_{m\geq 1} \frac{c_{2m}}{c_0}\, e^{-m_{2m} r}\right], \qquad m_{2m} = \frac{2\sqrt{m}}{\lambda}, \tag{61}$$

with the coefficient ratios

$$\frac{c_{2m}}{c_0} = \frac{1}{4^m}\binom{2m}{m} = 1,\ \frac{1}{2},\ \frac{3}{8},\ \frac{5}{16},\ \ldots \quad (m = 0, 1, 2, 3, \ldots). \tag{62}$$

Therefore the leading deviation from Coulomb at large r is

$$A_0(r) = \frac{\mu_0^{\rm eff} q_\star}{4\pi r}\left[1 + \frac{1}{2} e^{-2r/\lambda} + \mathcal{O}\!\left(e^{-2\sqrt{2}\,r/\lambda}\right)\right]. \tag{63}$$

The corresponding radial electric field is $E_r(r) = -\partial_r A_0(r)$, so the zero mode gives the usual $1/r^2$ falloff and the KK corrections are exponentially suppressed for $r \gg \lambda$.

Interpretation. The confinement width λ sets the crossover scale: for brane separations $r \gg \lambda$, the zero mode dominates and the theory is experimentally indistinguishable from Maxwell to extremely high accuracy; for $r \sim \lambda$, the tower becomes relevant and predicts specific, computable departures from the pure Coulomb law. In section 7 we extend this discussion from static sources to moving defect branches, where each KK mode propagates with its own covariant retarded Green's function and the full response remains brane-Lorentz covariant. Equivalently, after canonical normalization one may rewrite the same brane field in terms of $q_{\rm eff}$, but in either normalization the Yukawa tower modifies the field of a fixed topological charge branch; it does not imply that electric charge varies with circulation, throat radius, or breathing.

7 Moving sources and retarded structure

The preceding section established that the localized Maxwell sector reduces, for a brane-localized source, to a massless $3+1$ mode plus a discrete tower of massive KK excitations with computable couplings. Here we explain how this structure extends from static sources to fully time-dependent (moving) sources, and why the resulting brane response is causal and brane-Lorentz covariant. The key point is that each KK mode obeys a standard covariant wave equation in $3+1$ with a retarded Green's function that depends only on Lorentz invariants; summing the tower preserves that structure.

Mode-by-mode brane equations. In axial gauge with a brane-localized conserved current $J^\mu(x,w) = J_b^\mu(x)\delta(w)$, the KK decomposition $A_\mu(x,w) = \sum_n a_\mu^{(n)}(x) f_n(w)$ together with the orthogonality of the f_n under the $Z(w)$-weighted inner product yields a decoupled set of $3+1$ equations for the mode amplitudes $a_\mu^{(n)}(x)$. For conserved brane sources (so that longitudinal pieces can be gauged away in the usual way), each mode obeys a covariant Klein–Gordon-type operator with mass m_n,

$$\left(\Box_4 + m_n^2\right) a_\mu^{(n)}(x) = \mu_0\, c_n\, J_{b\,\mu}(x), \qquad \Box_4 \equiv \partial_t^2 - \nabla^2, \tag{64}$$

where the masses $m_n^2 = 2n/\lambda^2$ and couplings c_n were given in equations (52), (55) and (57). The zero mode $n=0$ is massless and reproduces standard Maxwell propagation on the brane (with effective coupling $\mu_0^{\rm eff} = \mu_0 c_0$), while the massive modes propagate as covariant massive vector excitations.

Retarded Green functions. Let $G_{\rm ret}^{(m)}(x-x')$ be the retarded Green function for the scalar operator $\Box_4 + m^2$,

$$(\Box_4 + m^2)\, G_{\rm ret}^{(m)}(x-x') = \delta^{(4)}(x-x'), \qquad G_{\rm ret}^{(m)}(x-x') = 0 \;\text{ for } t < t'. \tag{65}$$

Then the retarded solution of equation (64) is

$$a_\mu^{(n)}(x) = \mu_0\, c_n \int \mathrm{d}^4x'\; G_{\rm ret}^{(m_n)}(x-x')\, J_{b\,\mu}(x'). \tag{66}$$

Equivalently, in Fourier space the retarded prescription gives

$$G_{\rm ret}^{(m)}(x) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{\mathrm{e}^{-\mathrm{i}k\cdot x}}{k^2 + m^2 + \mathrm{i}\epsilon\, k^0}, \qquad k^2 \equiv \eta^{\mu\nu} k_\mu k_\nu = -\omega^2 + \mathbf{k}^2, \tag{67}$$

where the $\mathrm{i}\epsilon k^0$ term encodes the retarded boundary condition (and is invariant under proper orthochronous brane Lorentz transformations). The effective brane kernel is therefore a KK-weighted sum,

$$G_{\rm ret}^{\rm eff}(x) \equiv \sum_{n\geq 0} c_n\, G_{\rm ret}^{(m_n)}(x), \qquad a_\mu(x) = \mu_0 \int \mathrm{d}^4x'\; G_{\rm ret}^{\rm eff}(x-x')\, J_{b\,\mu}(x'), \tag{68}$$

and, in momentum space,

$$D_{\rm ret}^{\rm eff}(k) = \sum_{n\geq 0} \frac{c_n}{k^2 + m_n^2 + \mathrm{i}\epsilon\, k^0}. \tag{69}$$

This makes transparent the structural statement checked in the symbolic suite: the brane-to-brane response depends on brane momenta only through the Lorentz scalar k^2 (together with the standard retarded prescription), so there is no preferred-frame artifact in the brane dynamics.

Explicit causal support and "tails." In $3+1$, the massless retarded Green function is supported on the light cone,

$$G_{\text{ret}}^{(0)}(t,r) = \theta(t)\,\frac{\delta(t-r)}{4\pi r}, \qquad r \equiv |\mathbf{x}|, \tag{70}$$

while the massive retarded Green function has support on and *inside* the light cone (a "tail"):

$$G_{\text{ret}}^{(m)}(t,r) = \theta(t)\left[\frac{\delta(t-r)}{4\pi r} - \theta(t-r)\,\frac{m}{4\pi}\,\frac{J_1\!\left(m\sqrt{t^2-r^2}\right)}{\sqrt{t^2-r^2}}\right], \tag{71}$$

where J_1 is the Bessel function of the first kind. The wavefront remains lightlike (the $\delta(t-r)$ term), while massive modes contribute additional causal response inside the forward light cone. Consequently, the full KK-summed response equation (68) is manifestly causal.

Moving defect branches. For a moving defect on a fixed charge branch $q_\star$ moving on a brane worldline $z^\mu(\tau)$ with four-velocity $u^\mu = \mathrm{d}z^\mu/\mathrm{d}\tau$, the conserved current can be written as

$$J_b^\mu(x) = q_\star \int \mathrm{d}\tau\; u^\mu(\tau)\,\delta^{(4)}(x - z(\tau)). \tag{72}$$

Substituting equation (72) into equation (66) and summing modes gives

$$a_\mu(x) = \mu_0 q_\star \sum_{n\geq 0} c_n \int \mathrm{d}\tau\; u_\mu(\tau)\, G_{\text{ret}}^{(m_n)}(x - z(\tau)). \tag{73}$$

The $n=0$ term reproduces the usual retarded Maxwell potential (the Liénard–Wiechert structure), while the $n \geq 1$ modes add Yukawa-suppressed and tail-type corrections governed by the KK masses and brane couplings. In the far-field regime $r \gg \lambda$, these corrections are exponentially small, in accordance with equation (63). The worldline current therefore describes a moving fixed defect branch rather than an arbitrary continuously adjustable electric charge.

Brane Lorentz covariance. Two complementary viewpoints make the covariance manifest:

1. *Momentum-space:* the response kernel equation (69) depends on brane momenta only through k^2 (a Lorentz scalar), and the retarded prescription is preserved under proper orthochronous boosts. Thus the resulting fields transform correctly for all time-dependent sources.

2. *Position-space:* the Green functions equation (71) depend on separations only through r, t, and the invariant interval $t^2 - r^2$, together with causal step functions. Each KK mode therefore yields a brane-covariant retarded solution, and the KK sum preserves that property.

This completes the "no preferred-frame" argument for moving sources: once the EM sector is localized by $Z(w)$ without breaking brane Poincaré symmetry, the brane-effective dynamics are fully Lorentz covariant, with calculable deviations from pure Maxwell behavior controlled by the KK tower.

8 Matter current conservation and Maxwell consistency

The localized Maxwell equations equation (15) require a conserved source in order to be compatible with the gauge condition and the divergence identity equation (20). In a complete theory one should not impose source conservation by hand: it must follow from the matter dynamics and its symmetries. In this section we show that a minimal brane matter model with $U(1)$ phase symmetry produces a conserved Noether current (on shell), and that this current is precisely the object that can be used as the brane source for the localized Maxwell sector. This closes the logical loop:

$$\text{(matter symmetry)} \;\Rightarrow\; \partial_\mu j^\mu = 0 \;\Rightarrow\; \partial_M J_\psi^M = 0 \;\Rightarrow\; \text{Maxwell consistency.}$$

8.1 Minimal brane matter model and Noether current

We consider a minimal brane field $\psi(t, \mathbf{x})$ with a standard Schrödinger/Gross–Pitaevskii-type Lagrangian density and minimal gauge coupling. A representative form (sufficient for the current-conservation identity checked in the referee suite) is

$$\mathcal{L}_\psi = \frac{\mathrm{i}\hbar}{2}(\psi^* D_t \psi - \psi (D_t \psi)^*) - \frac{\hbar^2}{2m}(D_i \psi)^*(D_i \psi) - U(|\psi|^2), \tag{74}$$

where U is an arbitrary real function of $|\psi|^2$ and the covariant derivatives are

$$D_t \equiv \partial_t + \frac{\mathrm{i}q_\star}{\hbar} A_0, \qquad D_i \equiv \partial_i - \frac{\mathrm{i}q_\star}{\hbar} A_i, \tag{75}$$

with fixed defect-branch label $q_\star = \eta_Q e_\star$.

The matter sector is invariant under the local $U(1)$ transformation

$$A_0 \mapsto A_0 - \partial_t \chi, \qquad A_i \mapsto A_i + \partial_i \chi, \qquad \psi \mapsto \exp\left(\frac{\mathrm{i}q_\star}{\hbar}\chi\right)\psi, \tag{76}$$

which ensures that $D_\mu \psi$ transforms covariantly and $\mathcal{L}_\psi$ is gauge invariant (up to total derivatives associated with the time-derivative term). In particular, restricting χ to a constant recovers a global phase symmetry, and Noether's theorem yields a conserved current.

A convenient way to extract the current (and the approach used in the symbolic suite) is to probe a *local* phase variation $\psi \to e^{\mathrm{i}\epsilon(x)}\psi$, $\psi^* \to e^{-\mathrm{i}\epsilon(x)}\psi^*$ and expand to first order in ϵ. One finds a current j^μ on the brane with time component

$$j^0 = |\psi|^2, \tag{77}$$

and spatial components

$$j^i = \frac{\hbar}{2mi}\Big(\psi^* D^i \psi - (D^i \psi)^* \psi\Big) = \frac{\hbar}{m}\,\mathrm{Im}\Big(\psi^* D^i \psi\Big), \tag{78}$$

which is manifestly gauge covariant. The corresponding *charge current* is

$$J_\psi^\mu \equiv q_\star\, j^\mu. \tag{79}$$

This minimal brane matter model is used here as a representative source-consistency closure, not as a claim that the entire ambient vacuum condensate is uniformly charged. In the defect reading, ψ may be treated as a localized, unit-normalized defect profile carrying fixed total charge $q_\star$, while any neutralizing background is encoded separately in J^0_{ext}. In the full coupled theory, J_ψ^μ is the dynamical source generated by minimal coupling, while the explicit Maxwell action term carries only J^μ_{ext}.

8.2 Off-shell identity and linkage to Maxwell divergence

The local phase-probe method yields not only the explicit form of the Noether current but also an *off-shell* identity (Noether identity) relating its divergence to the matter Euler–Lagrange expressions. Denote the brane Euler–Lagrange operators by

$$\mathrm{EL}_\psi \equiv \frac{\partial \mathcal{L}_\psi}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}_\psi}{\partial(\partial_\mu \psi)} \right), \qquad \mathrm{EL}_{\psi^*} \equiv \frac{\partial \mathcal{L}_\psi}{\partial \psi^*} - \partial_\mu \left(\frac{\partial \mathcal{L}_\psi}{\partial(\partial_\mu \psi^*)} \right). \tag{80}$$

Then the phase symmetry implies the identity

$$\partial_\mu j^\mu + \frac{\mathrm{i}}{\hbar} \Big(\psi^* \mathrm{EL}_{\psi^*} - \psi\, \mathrm{EL}_\psi \Big) = 0. \tag{81}$$

On shell, $\mathrm{EL}_\psi = \mathrm{EL}_{\psi^*} = 0$, this reduces to the usual continuity equation

$$\partial_\mu j^\mu = 0, \qquad \text{equivalently} \qquad \partial_\mu J^\mu_\psi = 0. \tag{82}$$

The referee suite verifies equation (81) explicitly for the minimal matter model used there (including the corrected sign convention in the identity residual).

To embed the matter source into the localized 4+1 Maxwell sector, we use the brane-localized current model

$$J^\mu(x, w) = J^\mu_\psi(x)\, \delta(w), \qquad J^w(x, w) = 0, \tag{83}$$

as in section 5.2. Since $\delta(w)$ depends only on w, the bulk divergence reduces distributionally to

$$\partial_M J^M = \partial_\mu (J^\mu_\psi \delta(w)) + \partial_w(0) = \delta(w)\, \partial_\mu J^\mu_\psi. \tag{84}$$

Therefore, the on-shell brane conservation law equation (82) implies bulk conservation $\partial_M J^M = 0$. This is exactly the condition required by Maxwell consistency: inserting $\partial_M J^M = 0$ into the divergence identity equation (20) shows that in Lorenz gauge ($\partial \cdot A = 0$) the bulk equations are mutually compatible and the gauge-fixed operator closes without contradiction.

In summary, the same symmetry (phase invariance) that makes the matter sector well-posed also supplies the conserved source required by the localized Maxwell sector. In the controlled brane limit, this guarantees that the effective $3+1$ Maxwell equations are sourced by a current that is conserved by construction, rather than by assumption.

9 Discussion: validity regime, deviations, and falsifiable tests

The localized $4+1$ Maxwell sector yields ordinary $3+1$ electromagnetism on the brane only in a controlled regime. Outside that regime, the theory makes specific, quantitative predictions: Yukawa-suppressed corrections to Coulomb, a discrete set of massive response channels with threshold behavior in frequency, and (for time-dependent sources) causal "tail" terms inside the light cone. Because the correction structure is fixed by the localization profile $Z(w)$, these deviations are falsifiable rather than tunable.

9.1 When the $3+1$ Maxwell limit holds

The exact integrated brane equation equation (38) shows that the reduction to standard Maxwell is controlled by (i) the dominance of a w-independent zero mode in the field configuration and (ii) the absence of transverse current. A sufficient set of conditions for the Maxwell regime is:

1. **Zero-mode dominance:** the brane components are approximately w-independent within the support of $Z(w)$, $A_\mu(x, w) \simeq a_\mu(x)$ and $\partial_w A_\mu \simeq 0$, as in equation (39). Equivalently, the typical brane length/time scales are long compared to the confinement width:

$$r \gg \lambda, \qquad \omega \ll \frac{2}{\lambda}, \tag{85}$$

so that KK excitations are either not sourced efficiently or remain evanescent.

2. **No transverse source flux:** $J^w = 0$, so the $N = w$ Maxwell equation equation (43) is consistent with axial gauge and the zero-mode limit.

3. **Localized boundary behavior:** $Z(w)F^{w\nu}$ decays sufficiently fast as $|w| \to \infty$ so that the boundary term in equation (36) vanishes (automatic for the localized configurations considered here).

4. **Conserved brane current:** $\partial_\mu J_b^\mu = 0$, which follows from the matter dynamics via the Noether identity equation (81).

Under these conditions the theory reduces to the familiar 3+1 Maxwell equation equation (41) with effective coupling $\mu_0^{\mathrm{eff}} = \mu_0/Z_{\mathrm{int}}$; a convenient gauge such as $\partial_\mu a^\mu = 0$ may then be chosen within the reduced theory, and all gauge-invariant brane observables coincide with Maxwell to high accuracy.

In the updated charge ontology, this far-field Maxwell limit describes the brane field of a fixed defect branch with sign η_Q and effective brane coupling $q_{\mathrm{eff}} = q_\star/\sqrt{Z_{\mathrm{int}}}$. What is being suppressed here is the mixed core sector $(A_w, J^w, F_{\mu w})$, not the underlying puncture orientation itself.

9.2 Leading deviation structure

The Gaussian profile fixes the KK spectrum and brane couplings, yielding a rigid deviation pattern rather than an arbitrary modification. For static sources the brane potential takes the KK-summed form equation (60); separating the zero mode gives the leading correction equation (63),

$$A_0(r) = \frac{\mu_0^{\mathrm{eff}} q_\star}{4\pi r}\left[1 + \frac{1}{2}\,\mathrm{e}^{-2r/\lambda} + \mathcal{O}\Big(\mathrm{e}^{-2\sqrt{2}\,r/\lambda}\Big)\right]. \tag{86}$$

Two points are worth emphasizing:

- The *range* of the leading deviation is fixed by the first brane-coupled mass $m_2 = 2/\lambda$ (odd modes do not couple; equation (56)).
- The *amplitude* of the leading Yukawa term is also fixed (here 1/2) by the Gaussian mode structure (equation (62)). In this sense the Gaussian localization makes a parameter-light, falsifiable prediction: once λ is chosen, the leading correction is determined.

These Yukawa corrections modify the field profile of a fixed topological charge branch. They do not imply any dependence of electric charge on circulation, throat radius, or geometry breathing.

For time-dependent sources, each KK mode obeys a covariant massive wave equation equation (64) with retarded solution equation (66). Below the first brane-coupled threshold $\omega < m_2$, massive modes contribute only near-field/evanescent corrections and the response is extremely close to Maxwell at distances $r \gg \lambda$. Above threshold, $\omega > m_2$, additional propagating channels open with dispersion

$$\omega^2 = k^2 + m_n^2, \tag{87}$$

and group velocity $v_g = k/\omega < 1$. This predicts frequency-dependent response (and, in principle, additional polarization content) controlled by the discrete tower $m_n^2 = 2n/\lambda^2$ and couplings c_n.

9.3 Falsifiable tests and observational handles

The localization sector can be tested in several complementary ways. The discussion below is organized by which assumption in the controlled reduction is being probed.

(i) Precision tests of Coulomb's law (static). The cleanest signature is a Yukawa departure with *fixed coefficient pattern* set by $Z(w)$. For Gaussian localization the leading correction has coefficient $1/2$ and range $\lambda/2$ in the exponent (equation (63)). Any experimental bound on Yukawa deviations from $1/r$ behavior can be translated directly into a bound on λ. Conversely, a detected deviation would overconstrain the model because higher-mode coefficients are not free: the subleading terms are predicted to follow equation (62) with masses $\{2\sqrt{m}/\lambda\}_{m\geq 1}$.

(ii) Threshold behavior and dispersion (time-dependent). For driven sources at frequency ω, the theory predicts a qualitative change in response when ω crosses the first coupled KK mass $m_2 = 2/\lambda$. Below threshold, KK contributions are evanescent; above threshold they propagate with subluminal group velocity. If λ were large enough to place m_2 in an accessible band, this would appear as a sharp onset of dispersive components or additional attenuation channels. If no such effect is seen up to some frequency, one obtains a lower bound on m_2, hence an upper bound on λ.

(iii) Retarded "tail" structure (time-domain). Massive modes produce causal tails inside the forward light cone (equation (71)), implying that a sharp electromagnetic pulse sourced on the brane can generate a small, model-predicted after-response beyond the light-cone wavefront. Detecting (or constraining) such tails provides a direct probe of massive KK content distinct from static-force measurements.

(iv) Charge conservation and possible leakage signatures. The controlled reduction requires $J^w = 0$ and yields bulk conservation $\partial_M J^M = 0$ when the brane matter equations enforce $\partial_\mu J_b^\mu = 0$ (equation (82)). Any empirical signature of apparent charge nonconservation on the brane (or of systematic violations of the Maxwell divergence identity) would falsify the source model equation (33) and force the inclusion of transverse current or additional degrees of freedom. Conversely, the framework offers a precise place to parametrize such effects: $J^w \neq 0$ is not "forbidden," but it moves one outside the Maxwell regime and predicts accompanying modifications in the $N = w$ equation equation (43).

(v) Profile-shape tests beyond the Gaussian. While this paper adopts a Gaussian $Z(w)$ for analytic control, the general framework depends on $Z(w)$ only through the Sturm–Liouville problem equation (50) and the resulting tower $\{m_n, c_n\}$. Different profiles produce different discrete spectra and coupling patterns. Thus, experimental constraints can be interpreted either as bounds on λ *within* the Gaussian class, or more broadly as constraints on the admissible profile shapes that yield the observed near-perfect Maxwell behavior on the brane.

9.4 Relation to the unified model and next derivations

Within the broader unified program, the localized Maxwell sector plays two roles: (i) it supplies a genuine gauge dynamics on the brane (including radiation and Lorentz-force coupling), and (ii) it provides a controlled place to quantify deviations (KK tails, Yukawa corrections) that can be linked to the transverse structure of the unified geometry. A natural next step is to derive or constrain $Z(w)$ and λ from the throat/defect microphysics rather than treating them as phenomenological response data. The referee suite provided with this paper is designed to make that future step modular: any proposed $Z(w)$ can be dropped into the same symbolic pipeline to produce the corresponding μ_0^{eff}, q_{eff}, KK spectrum, and brane deviation pattern. The mixed $(A_w, J^w, F_{\mu w})$ sector remains the natural place for future defect-core twist/spin dynamics, but that microphysics is outside the scope of the present Maxwell-localization paper.

10 Conclusion

We isolated the gauge-field sector of the unified $4+1$ framework and showed that ordinary $3+1$ Maxwell theory on the brane emerges as a controlled, script-verified limit of a localized 5D Maxwell action. The core ingredients are: (i) a transverse localization profile $Z(w)$ multiplying the Maxwell kinetic term, (ii) standard gauge structure (with an explicit gauge-fixing term used here to make operator inversion and consistency checks transparent), and (iii) a brane-localized conserved source.

Starting from the localized action equation (8), variation with respect to A_N yields the bulk Euler–Lagrange equations equation (15). The same framework implies the homogeneous Maxwell equations via the Bianchi identities equation (17), and it enforces a precise divergence consistency identity equation (20) relating the gauge condition to current conservation. We also made explicit the gauge-variation structure of the source coupling equation (24) and the correct transformation-type assignments (covector A_μ, vector J^μ) required for clean brane Lorentz-covariance checks.

Under axial gauge and a brane-dominant zero-mode ansatz, integrating the bulk equations over w reduces the dynamics to standard $3+1$ Maxwell theory with a renormalized coupling

$$\mu_0^{\text{eff}} = \frac{\mu_0}{Z_{\text{int}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w)\,\mathrm{d}w. \tag{88}$$

For the Gaussian profile $Z(w) = \exp(-w^2/\lambda^2)$, $Z_{\text{int}} = \lambda\sqrt{\pi}$ and $\mu_0^{\text{eff}} = \mu_0/(\lambda\sqrt{\pi})$. Beyond this Maxwell regime, the same localization mechanism predicts a discrete KK tower governed by the Sturm–Liouville problem equation (50). For Gaussian Z, the spectrum is $m_n^2 = 2n/\lambda^2$ with an even-mode coupling selection rule ($c_{2m+1} = 0$), yielding a Coulomb potential plus Yukawa-suppressed corrections whose leading term has fixed coefficient $1/2$ and range set by λ (equation (63)). For time-dependent sources, each KK mode propagates with a covariant retarded Green function, so the full brane response remains causal and brane-Lorentz covariant, with calculable dispersive/tail corrections (section 7).

Finally, we closed the consistency loop on the source side. A minimal brane matter model with $U(1)$ phase symmetry yields a Noether current j^μ with $j^0 = |\psi|^2$ and gauge-covariant spatial components equation (78), together with the off-shell identity equation (81) implying $\partial_\mu j^\mu = 0$ on shell. Embedding the associated charge current as a brane-localized bulk source $J^\mu = J^\mu_\psi \delta(w)$, $J^w = 0$ ensures $\partial_M J^M = 0$, precisely the condition required by the Maxwell divergence identity and by gauge consistency.

In summary, the localized Maxwell sector provides a complete, falsifiable mechanism for obtaining brane electromagnetism from the unified $4+1$ model: standard Maxwell dynamics emerge at long distances and low frequencies, while well-structured deviations (Yukawa corrections, discrete thresholds, and causal tails) appear when transverse excitations are relevant. The accompanying symbolic referee suite was designed to make these claims reproducible and to enable direct future substitutions of alternative localization profiles $Z(w)$ derived from the underlying defect/throat microphysics. In this updated ontology, the electric branch sign is fixed by η_Q, the observable brane charge strength is thickness-controlled through Z_{int} (equivalently q_{eff}), and the zero-mode reduction suppresses rather than eliminates the mixed core sector. This paper derives the localized Maxwell dynamics and the thickness-controlled brane charge strength; it does not derive spin or the microscopic mixed-twist defect-core dynamics.

A Referee verification suite summary (Wolfram Language)

This paper is accompanied by a compact Wolfram Language (WL) verification suite that checks the principal identities and reductions used in the main text. The suite is designed to be referee-friendly: each script prints the exact symbolic Lagrangian it varies (when applicable), converts

`EulerEquations` output to expressions before comparison (to avoid `Equal[...]` mismatches), and tests all nontrivial identities by simplifying the *residual* to zero under the declared conventions. In particular, the suite locks in the sign conventions and transformation types used throughout the paper: metric $\eta_{MN} = \mathrm{diag}(-1, 1, 1, 1, 1)$, field strength $F_{MN} = \partial_M A_N - \partial_N A_M$, source coupling written in compact Maxwell-only form as $\mathcal{L}_{\text{int}} = -A_M J^M$ with J^M contravariant and A_M covariant, and Gaussian localization $Z(w) = \exp(-w^2/\lambda^2)$. In the fully coupled embedding, the explicit Maxwell source term is instead $-A_M J^M_{\text{ext}}$, while the dynamical matter current J^M_ψ is generated separately by minimal coupling so that $J^M_{\text{tot}} = J^M_\psi + J^M_{\text{ext}}$.

(A) `maxwell_from_4d.wl`: action variation, EOM, Bianchi, divergence identity, brane reduction. This is the core referee harness. It defines the localized Maxwell action with Gaussian $Z(w)$, includes the Lorenz-type gauge-fixing term, and varies with respect to A_N. It verifies: (i) the full $4+1$ Euler–Lagrange equations match $\partial_M(ZF^{MN}) + (1/\xi)\partial^N(\partial\cdot A) = \mu_0 J^N$ for all components $N \in \{t, x, y, z, w\}$ (residuals simplify to 0); (ii) the Bianchi identities $\partial_{[L}F_{MN]} = 0$ hold identically for all index triples (residuals simplify to 0); (iii) the divergence consistency identity $\partial_N(\mathrm{EOM}^N) = \xi^{-1}\Box(\partial\cdot A) - \mu_0\,\partial_N J^N$ holds exactly (residual 0); and (iv) under a controlled brane/zero-mode reduction ($A_w = 0$, $\partial_w A_\mu = 0$), the integrated equations reproduce standard $3+1$ Maxwell with $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}} = \mu_0/(\lambda\sqrt{\pi})$. The same harness also prints the updated charge bookkeeping $q_\star = \eta_Q e_\star$, $J^M_{\text{tot}} = J^M_\psi + J^M_{\text{ext}}$, and the equivalent canonical brane-coupling relation $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$.

(B) `maxwell_symmetry_checks.wl`: gauge variation and brane Lorentz invariance. This script checks the symmetry structure emphasized in section 4. It verifies the gauge variation of the interaction term, $\delta\mathcal{L}_{\text{int}} = -\partial_M(\chi J^M) + \chi\,\partial_M J^M$, and the shift $\delta(\partial\cdot A) = \Box\chi$ implied by $\delta A_M = \partial_M\chi$. It also verifies brane Lorentz invariance by implementing the correct transformation types: A_μ as a covector and J^μ as a vector. In addition, it checks the Paper VII matter-gauge convention $A_0 \mapsto A_0 - \partial_t\chi$, $A_i \mapsto A_i + \partial_i\chi$, $\psi \mapsto e^{\mathrm{i}q_\star\chi/\hbar}\psi$, confirming that the corresponding covariant derivatives transform homogeneously. With this bookkeeping, it confirms invariance of $F_{\mu\nu}F^{\mu\nu}$ and $A_\mu J^\mu$ under representative boosts (residuals simplify to 0).

(C) `maxwell_kk_coulomb_propagator.wl`: Gaussian KK spectrum, couplings, Coulomb+Yukawa, and $D_{\text{eff}}(k^2)$. This script establishes the transverse Sturm–Liouville eigenproblem $-\frac{\mathrm{d}}{\mathrm{d}w}(Zf') = m^2 Zf$ for Gaussian Z and verifies the Hermite solutions $f_n(w) = H_n(w/\lambda)$ with spectrum $m_n^2 = 2n/\lambda^2$. It validates the mode norms $\|f_n\|^2 = \lambda\sqrt{\pi}\,2^n n!$ by integer spot-checks and then uses the closed form to compute brane couplings $c_n = f_n(0)^2/\|f_n\|^2$, confirming the parity selection rule $c_{2m+1} = 0$ and the even-mode formula $\frac{c_{2m}}{c_0} = \frac{1}{4^m}\binom{2m}{m}$. It then derives the Coulomb zero-mode potential and the KK Yukawa corrections for a fixed defect branch labeled by $q_\star$, records the equivalent canonical coupling $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$, and notes explicitly that odd transverse structure can decouple from centered brane sources without being absent microscopically. Finally, it checks that truncated Euclidean (and Minkowski) effective propagators depend only on the brane Lorentz scalar k^2.

(D) `maxwell_axial_gauge_brane_sources.wl`: axial gauge reachability, residual $3+1$ gauge, $\delta(w)$ sources, and integrated EOM. This script demonstrates explicit axial gauge reachability via the parameter $\chi_{\text{Ax}}(w) = -\int_0^w A_w\,\mathrm{d}w'$, and shows that the residual gauge freedom is w-independent (standard $3+1$ gauge parameter). It implements a brane-localized current $J^\mu = J^\mu_b(x)\delta(w)$, $J^w = 0$, and verifies: (i) the $N = w$ Maxwell equation is consistent with axial gauge and the zero-mode ansatz; and (ii) integrating the $N = \nu$ equations over w yields precisely the $3+1$ Maxwell form with $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$. The harness also records the interpretive caveat used in the paper: $J^w = 0$ is a controlled far-field Maxwell-limit condition, not a claim that the microscopic defect core lacks mixed $(A_w, J^w, F_{\mu w})$ structure.

(E) `maxwell_moving_charges_propagator.wl`: covariant response for time-dependent sources. This script provides a complementary momentum-space verification of the statements in section 7. It demonstrates explicitly that k^2 is brane boost invariant and that the KK-truncated effective propagator depends only on k^2, and it supplies the corresponding Minkowski $i\epsilon$ prescriptions for retarded structure mode-by-mode. In the static limit, it reproduces the Coulomb+Yukawa structure for a fixed defect branch labeled by $q_\star$; more generally it supports the statement that moving defect branches do not introduce preferred-frame artifacts in the brane response.

(F) `matter_current_conservation.wl`: Noether current and off-shell identity. This script derives the brane matter Noether current by probing a local phase variation $\epsilon(t, x, y, z)$ in a minimally coupled matter Lagrangian of the form summarized in equation (74). It verifies: (i) constant phase transformations leave $\mathcal{L}_\psi$ invariant; (ii) the extracted number current satisfies $j^0 = |\psi|^2$ and the expected Paper VII gauge-covariant spatial form; and (iii) the off-shell Noether identity $\partial_\mu j^\mu + \frac{i}{\hbar}(\psi^* \mathrm{EL}_{\psi^*} - \psi\, \mathrm{EL}_\psi) = 0$ holds exactly (residual simplifies to 0) with the fixed sign convention used in this paper. It then records the Maxwell source as $J^\mu_\psi = q_\star j^\mu$, with $q_\star = \eta_Q e_\star$, matching the localized-defect / background-split bookkeeping used in the main text.

Reproducibility note. All scripts are intended to be run in a standard WL kernel (e.g. via `wolframscript -file <file.wl>`). Each prints pass/fail messages and (where appropriate) the residual expressions before simplification. The suite is structured so that replacing the profile $Z(w)$ by an alternative localization ansatz requires changes in only one place, after which the same EOM, identity, KK-spectral, reduction, and charge-normalization checks can be re-run verbatim.

B Sign conventions and common pitfalls

This appendix records the sign, index, and transformation conventions used in the paper and highlights the few implementation pitfalls that historically cause spurious discrepancies (especially in component-based symbolic checks). The goal is to make it easy for readers and referees to reproduce results without re-deriving the bookkeeping.

B.1 Frozen conventions used throughout

Coordinates and metric. We use bulk coordinates $x^M = (t, x, y, z, w)$ with $\eta_{MN} = \mathrm{diag}(-1, 1, 1, 1, 1)$. The bulk d'Alembertian is $\Box = \eta^{MN}\partial_M\partial_N = -\partial_t^2 + \nabla^2 + \partial_w^2$. Brane indices are $\mu, \nu \in \{0, 1, 2, 3\}$ and spatial brane indices $i, j \in \{1, 2, 3\}$.

Field strength. The field strength is defined by

$$F_{MN} = \partial_M A_N - \partial_N A_M, \qquad F^{MN} = \eta^{MP}\eta^{NQ}F_{PQ}. \tag{89}$$

With these definitions, the Bianchi identities $\partial_{[L}F_{MN]} = 0$ hold identically.

Localization profile. The localization function $Z(w)$ multiplies the Maxwell kinetic term. In the main text we take the Gaussian $Z(w) = \exp(-w^2/\lambda^2)$, with finite integral $Z_{\mathrm{int}} = \int Z\, \mathrm{d}w = \lambda\sqrt{\pi}$.

Source coupling sign and index placement. The compact Maxwell-only source term is written as

$$\mathcal{L}_{\rm int} = -A_M J^M, \tag{90}$$

where J^M is treated as *contravariant* (a vector field), and A_M is *covariant* (a one-form). In the fully coupled embedding, however, the explicit Maxwell action carries only the external/background source, $-A_M J^M_{\rm ext}$, while the dynamical matter current J^M_ψ is generated by minimal coupling, so that $J^M_{\rm tot} = J^M_\psi + J^M_{\rm ext}$. With the compact convention, a gauge transformation $\delta A_M = \partial_M \chi$ yields

$$\delta\mathcal{L}_{\rm int} = -(\partial_M\chi)J^M = -\partial_M(\chi J^M) + \chi\,\partial_M J^M, \tag{91}$$

so gauge invariance of the coupling (up to boundary terms) is equivalent to $\partial_M J^M = 0$.

Charge labels and reduced coupling. The updated charge ontology distinguishes the fixed defect-branch sign $\eta_Q = \pm 1$ from the positive microscopic scale $e_\star > 0$, with $q_\star \equiv \eta_Q e_\star$. The canonically normalized brane coupling is then $q_{\rm eff} = q_\star/\sqrt{Z_{\rm int}} = \eta_Q\, e_{\rm eff}$, where $e_{\rm eff} \equiv e_\star/\sqrt{Z_{\rm int}}$. Thus $q_\star$ labels the microscopic defect branch, while $q_{\rm eff}$ measures the observable brane coupling after zero-mode normalization; neither sign nor magnitude is defined here by circulation or throat breathing.

Gauge fixing. We include a Lorenz-type gauge-fixing term $-(\partial{\cdot}A)^2/(2\xi\mu_0)$ with $\partial{\cdot}A \equiv \partial_M A^M$. This term explicitly breaks gauge invariance, as expected, but renders the operator invertible and makes the divergence identity and component checks unambiguous.

B.2 Lorentz transformations: covector vs vector

A persistent source of apparent "Lorentz non-invariance" in component checks is using the wrong transformation type for A_μ. In this paper:

- A_μ transforms as a *covector* (one-form): $A'_\mu(x') = (\Lambda^{-1})^\nu{}_\mu\, A_\nu(x)$.
- J^μ transforms as a *vector*: $J'^\mu(x') = \Lambda^\mu{}_\nu\, J^\nu(x)$.

With these assignments, $A_\mu J^\mu$ is a Lorentz scalar and $F_{\mu\nu}F^{\mu\nu}$ is invariant. Treating A_μ as a vector instead will generally produce spurious sign/index mismatches in invariance checks.

B.3 Brane reduction: what must vanish, and why

The controlled brane Maxwell limit relies on two closely related assumptions:

1. **Axial gauge and zero-mode dominance:** $A_w = 0$ and $\partial_w A_\mu \simeq 0$ over the localization region so that $F^{w\nu} = -\partial^w A^\nu \simeq 0$.
2. **No transverse source flux:** $J^w = 0$, consistent with the reduced $N = w$ equation equation (43).

If $J^w \neq 0$ or if significant w-dependence is present, the brane theory departs from pure Maxwell in a way controlled by the KK tower, as discussed in section 9. These conditions therefore define a controlled far-field sector: they suppress the mixed core channels $(A_w, J^w, F_{\mu w})$ without implying that such structure is absent from the microscopic defect ontology.

B.4 Implementation pitfalls in symbolic workflows (WL)

The accompanying WL suite is structured to avoid several common pitfalls that can make correct physics look incorrect:

(i) `EulerEquations` returns `Equal[...]`. In WL, `EulerEquations` often returns equations as `lhs == rhs`. When checking equality against a target expression, it is safer to convert to a single expression via `lhs - rhs` before simplifying. Directly simplifying a difference of `Equal` objects can yield misleading nonzero residuals.

(ii) Protected iterators and malformed replacement rules. Using symbolic iterators in `Table` (e.g. $\{a, b, c\}$ as if they were numeric ranges) can trigger protected-symbol errors. Likewise, constructing replacement rule lists indirectly (e.g. via nested `Join/Table` patterns) can silently produce malformed rule sets. The suite avoids these issues by using explicit index lists and clean, flat replacement-rule lists.

(iii) Symmetry checks require correct index raising/lowering. Sign mismatches often arise from mixing covariant and contravariant components without explicitly applying η_{MN} or η^{MN}. The suite always computes $A^M = \eta^{MN} A_N$ and $F^{MN} = \eta^{MP}\eta^{NQ}F_{PQ}$ before forming divergences or contractions.

(iv) Noether identity sign convention. The off-shell Noether identity used in equation (81) has a definite relative sign between $\partial_\mu j^\mu$ and the Euler–Lagrange combination $(\psi^* \mathrm{EL}_{\psi^*} - \psi\, \mathrm{EL}_\psi)$. Using the opposite sign will produce an apparent "failure" even if the extracted current is correct. The suite implements the identity in the form written in the main text.

(v) KK norms for symbolic mode number. Direct symbolic integration of Hermite norms for generic n can return `Indeterminate` in WL. The suite validates the closed form by spot-checking integer n and then uses the validated closed-form expression for general statements about c_n and the Yukawa tower.